

2 Complexity Theory (tg508)

- (a) Define the class BPP formally. [2 marks]
- (b) Show that the standard choice of error probability bound in the definition of BPP can be reduced to an exponentially small error. [4 marks]
- (c) Suppose we set the error probability bound to only require less than $1/2$ (i.e., success probability at least $1/2$) in the above definition. Would that change the resulting complexity class? [*Clarification:* you can assume $\text{NP} \not\subseteq \text{BPP}$] [6 marks]
- (d) Prove that $\text{BPP} \subseteq \text{P/Poly}$. [4 marks]
- (e) Is $\text{BPP} = \text{P/Poly}$? Justify your answer. [4 marks]